

Luke Liang

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EDUCATION

MS in Computer Science, Machine Learning Track
Columbia University

Expected Dec 2026
New York, NY

BS in Computer Science, BS in Data Science & Statistics (GPA: 3.86)
Miami University

Aug 2021 - May 2024
Oxford, OH

- Astronaut Scholarship (*national*), Choose Ohio First ML Scholarship, Summa Cum Laude, Honors College
- Trained at University of Tennessee (*National Science Foundation sponsored*) for ML and data science through the HPC at Oak Ridge national laboratory (*only student selected*)

PROFESSIONAL EXPERIENCE

Research Assistant
Miami University

Aug 2021 - Oct 2024
Oxford, OH

- Published 4 papers focused on ML and network science (*1 first author, 3 co-author*). Two first author manuscripts involving web applications and simulation under preparation
- Created a self-contained web-app project housing a simulation platform for researchers at the Centers for Disease Control and Prevention (CDC) to evaluate policy and intervention effectiveness on youth suicide prevention (**Python, Numba, NumPy, Docker, HPC**)
- Applied various ML models and several methods to augment image data to identify erroneous network simulation models with up to 95% accuracy in select cases (**Python, ML, TensorFlow, NumPy, pandas**)
- Developed a simulation of HIV spreading on a cell level with mutations and drug resistance to validate treatment plans (**Python, NumPy**)
- Contributed in a National Health Institute suicide modeling grant proposal

Teaching Assistant (Machine Learning & Algorithms)
Miami University

Aug 2022 - Dec 2023
Oxford, OH

- Led and published an extension of a previous research project by exploring data compression techniques and several ML models for identifying erroneous network simulations (**Python, ML**)
- Tutored students in Python, data structures, algorithms, and machine learning
- Provided feedback and graded exams, assignments, quizzes, and projects for students

Software Engineering Intern
JPMorgan Chase & Co.

Jun 2022 - Aug 2022
Chicago, IL

- Migrated internal software from AngularJS to ReactJS and deployed the updated framework across the bank's infrastructure (**ReactJS, Jenkins**)
- Won the Code for Good Hackathon 2022 with a web-based project to revamp a non-profit organization's website with a group of other interns (**Javascript, MySQL**)
- Integrated legacy pages with new internally developed APIs and updated technical documents

SKILLS

Python, Machine Learning (ML), NumPy, TensorFlow, Network Science, C++, Javascript, Numba, pandas, ReactJS, Java, Linux, Git, HTML/CSS, MySQL, HPC, Docker, R, SAS

LEADERSHIP EXPERIENCE

Chapter President
Tau Beta Pi - The Engineering Honor Society

May 2023 - May 2024
Oxford, OH

- Established chapter through student government to secure funding for social and service events
- Increased membership and community engagement by over 70% after inactivity due to COVID